

Density Matrices

→ Conditions → $\text{Tr}(\rho) = 1$

→ $\rho \geq 0$ → Positive Semidefiniteness

Trace of a Matrix

$$\text{Tr}(A) = \sum_i a_{ii}$$

$$\text{Tr}(aA + bB) = a \text{Tr}(A) + b \text{Tr}(B)$$

$$\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$$

Positive Semidefiniteness

① $\rho = M^\dagger M$ → There exists M

② $\rho = \rho^\dagger$ → All eigenvalues are

non-negative real numbers

③ $\langle \rho \rangle = \langle \psi | \rho | \psi \rangle \geq 0$ → For all $|\psi\rangle$

$$\alpha = \bar{\beta} \beta$$

$$\alpha = \bar{\alpha} \quad \text{and} \quad \alpha \geq 0$$

↓
Number
Analogy

→ Connection to Quantum State Vectors

$$\rho = |\psi\rangle\langle\psi| \rightarrow \text{Pure State}$$

$$|\psi\rangle = (\alpha_i)_{n \times 1}$$

$$i \in [0, n-1]$$

$$|\psi\rangle\langle\psi| =$$

Ans.

→ Global Phase Degeneracy is Eliminated

$$|\phi\rangle = e^{i\theta} |\psi\rangle$$

$$\rho = |\phi\rangle\langle\phi| = |\psi\rangle\langle\psi|$$

→ Probabilistic States

$$\rho = \sum_{a=0}^{n-1} p_a |a\rangle\langle a| = \begin{bmatrix} p_0 & & 0 \\ & \ddots & \\ 0 & & p_{n-1} \end{bmatrix}$$

→ Probabilistic
state

Spectral Theorem and Density Matrices

$$\rho = \sum_{k=0}^{n-1} p_k |\psi_k\rangle \langle \psi_k|$$

$p_k \rightarrow$ Eigenvalues of ρ

$|\psi_k\rangle \rightarrow$ Eigenvectors of ρ

$\{|\psi_0\rangle, \dots, |\psi_{n-1}\rangle\} \rightarrow$ Orthonormal Basis

$$\begin{aligned} \text{Tr}(\rho) &= \text{Tr} \left(\sum_{k=0}^{n-1} p_k |\psi_k\rangle \langle \psi_k| \right) = \sum_{k=0}^{n-1} p_k \text{Tr}(|\psi_k\rangle \langle \psi_k|) \\ &= \sum_{k=0}^{n-1} p_k \text{Tr}(\langle \psi_k | \psi_k \rangle) = \sum_{k=0}^{n-1} p_k = 1 \end{aligned}$$